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On Detecting Hallucinations in Vision-Language Model Generated Captions for Remote Sensing Imagery

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Eidesstattliche Erklärung

Hiermit erkläre ich an Eides statt, dass ich die vorliegende Arbeit selbstständig und ohne fremde Hilfe angefertigt habe. Sämtliche benutzten Informationsquellen sowie das Gedankengut Dritter wurden im Text als solche kenntlich gemacht und im Literaturverzeichnis angeführt. Die Arbeit wurde bisher nicht veröffentlicht und keiner Prüfungsbehörde vorgelegt.

Hereby I declare that I wrote this thesis myself with the help of no more than the mentioned literature and auxiliary means.

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Acknowledgements

Abstract

This template is intended to give an introduction of how to write diploma and master thesis.

Zusammenfassung

Da die meisten Leute an der TU Deutsch als Muttersprache haben, muss das Abstract zusätzlich auch in Deutsch geschrieben werden.

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List of Acronyms

3GPP	3rd Generation Partnership Project
AJAX	Asynchronous JavaScript and XML
API	Application Programming Interface
AS	Application Server
CSCF	Call Session Control Function
CSS	Cascading Stylesheets
DHTML	Dynamic HTML
DOM	Document Object Model
FOKUS	Fraunhofer Institut fuer offene Kommunikationssysteme
GUI	Graphical User Interface
GPS	Global Positioning System
GSM	Global System for Mobile Communication
HTML	Hypertext Markup Language
HSS	Home Subscriber Server
HTTP	Hypertext Transfer Protocol
I-CSCF	Interrogating-Call Session Control Function
IETF	Internet Engineering Task Force
IM	Instant Messaging
IMS	IP Multimedia Subsystem
IP	Internet Protocol
J2ME	Java Micro Edition
JDK	Java Developer Kit
JRE	Java Runtime Environment
JSON	JavaScript Object Notation
JSR	Java Specification Request
JVM	Java Virtual Machine
NGN	Next Generation Network

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1 Introduction

2 Related Work

2.1 Image Captioning

2.1.1 Overview and State-of-the-Art

2.1.2 Vision-Language Models

2.1.3 Remote Sensing Image Captioning

2.2 Hallucination

2.2.1 Hallucinations in Natural Language Generation

The term hallucination in the context of machine learning is commonly defined as “content that is nonsensical or unfaithful to the provided source content” [3]. In practice, “hallucination” is used to describe a vast array of different errors with different underlying mechanisms that result in similar symptoms. A few examples of errors commonly called hallucinations include: when language models are consistently wrong because of erroneous training data; when language models “lie” in pursuit of a reward; when a language model’s answer is sensitive to irrelevant details such as a random seed (defined as “confabulation”) [4]. Many more types of domain-specific hallucinations exist, such as object hallucinations and co-occurrence hallucinations common in Vision-Language image captioning [5]. The term has expanded to include anything generated by a foundation model that is not based in “fact” or “faithfulness”. Generated content is “faithful” if based in source knowledge and “factual” if based in world knowledge. However, many applications treat source input as “fact” making “faithfulness” and “factuality” interchangeable [6]. Thus, hallucinations can be categorized into “intrinsic hallucinations”, unfaithful content that contradicts source knowledge, and “extrinsic hallucinations”, unfaithful content that cannot be verified by source knowledge but might still be factual [6].

One of the main causes of hallucinations are data-related problems, such as when target data contains additional information not supported by the source information [6]. This source-reference divergence may be caused by heuristic data collection used to build large datasets. For instance, the authors of WIKIBIO [7] collected and utilized infoboxes from Wikipedia pages as source, and the first sentence of the same Wikipedia page as target. This resulted in source-reference divergence in 62% of cases, as most WIKIBIO first sentences contain additional information not supported by the corresponding infobox. Source-reference divergence may also be innate in some NLG tasks. For example, certain chat systems enhance their engagingness with relevant facts that are not present in the user input, history or provided knowledge source, leading to inevitable extrinsic hallucinations [6]. Another possible hallucination cause is the presence of duplicates in datasets, which may bias the model into repeating memorized sentences present in

duplicated examples. Hallucinations may also be caused by limited exposure to niche domain-specific data or a general lack of source-target pairs, both problems are prevalent in Remote Sensing image-captioning [8].

The second main cause for hallucinations are training-related issues that affect different components of neural models. Encoders, responsible for compressing input data into meaningful representations, can have defective comprehension abilities or learn wrong correlations in data, increasing the rate of hallucination. Decoders, responsible for transforming encoded representations into output text, can attend to the wrong part of the encoded input or introduce randomness through sampling strategies and increase hallucination probability. Large pre-trained models can also suffer from parametric knowledge bias, this can lead to the model prioritizing parametric knowledge over provided input [6].

2.2.2 Hallucinations in Vision-Language applications

Hallucination Types

In Vision-Language, text is generated based on visual input. Images are a fundamentally different medium to text, and as such, the hallucinations that occur in image captioning are different from those that occur in other NLG tasks. VLM-Hallucinations occur in two different settings: “judgment hallucinations” occur when the response to a user’s question contradicts the actual image, “description hallucinations” are failures to faithfully depict the contents of an image [1]. Both types of hallucination can be seen in Figure 2.1. Errors can be in different visual semantics, such as objects, attributes, relations, position and count. Examples of these errors are also depicted in Figure 2.1.

Object hallucinations happen when generated text includes objects that do not actually exist in the input image. Attribute hallucinations occur when generated text claims incorrectly assigns a quality, such as color, to an existing object in an image. A relation hallucination is when the relationship between two objects is incorrectly described, for example “the car is to the left of the bicycle” or “the cloud is casting a shadow on the beach”. Hallucinations due to count error are when the count of a specific, existing object in an image is wrong, for example, if a generated description states that there are 4 cars in an image when in reality there are only 3 [9].

2 Related Work

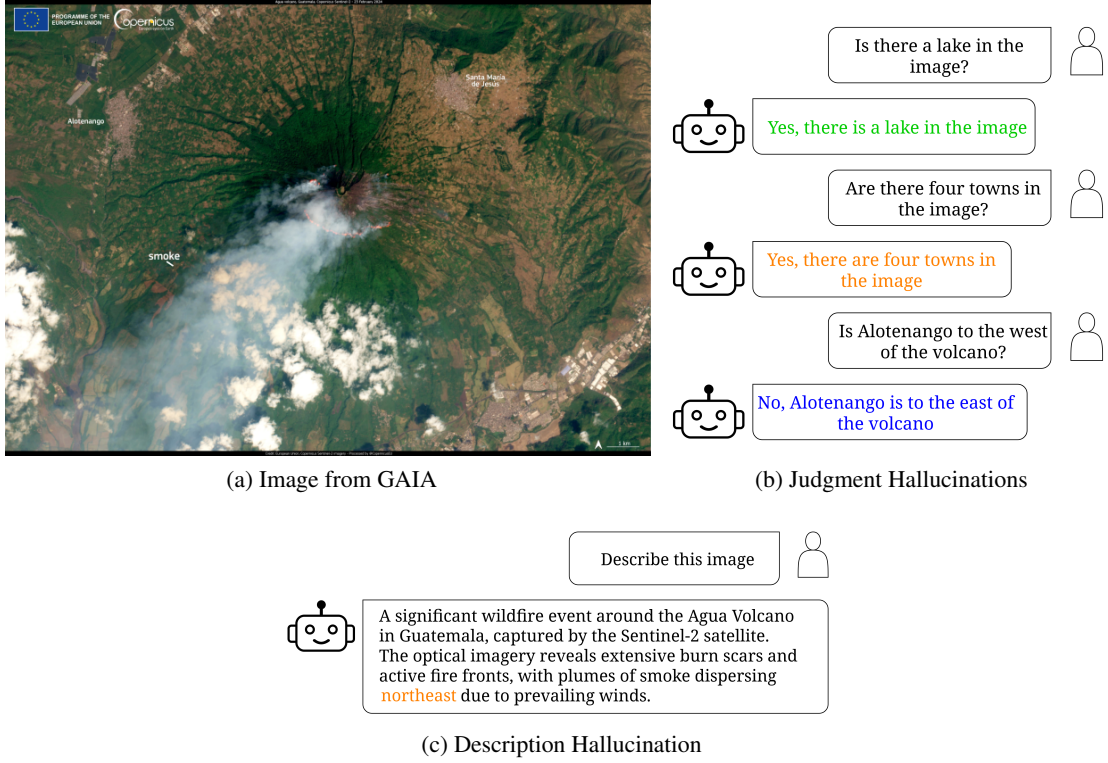


Figure 2.1: Examples of different types of hallucinations and visual semantic errors. In green: object hallucination; in orange: attribute hallucination; in blue: relation hallucination. Adapted from Liu et al. [1]. Image taken from GAIA dataset [2]

Hallucination Causes

Most of the literature on VLM-Hallucinations has focused on object hallucinations. The two primary causes of object hallucinations in LVLMs are statistical bias and language priors [10].

Statistical bias refers to a model’s tendency to rely on superficial object correlations and imbalanced distributions found in its vision-language training data. Due to statistical bias, models are more likely to hallucinate objects that are seen often during training. This also leads to co-occurrence hallucinations, models are more likely to hallucinate objects that frequently appear in the same images as other objects. For example, a model might incorrectly hallucinate a “fork” simply because it correctly identified a “dining table” in the image, or hallucinate a “parking lot” because it correctly identified a “shopping mall”.

Language priors are inherited by LVLMs from their language model component. Since LLMs are pre-trained on vast amounts of text-only data, they are designed to predict the next word based on linguistic probability rather than visual evidence. For example, Alayrac et al. [11] show that Large VLMs, when prompted with a question and image pair, hallucinate by producing answers that seem plausible given the text only, but contradict the contents of the given image. Additionally, visual uncertainty, such as unclear or distorted images, can exacerbate the reliance

on language priors. Leng et al. [10] illustrate this with an image of a “black banana”, as seen in Figure 2.2. While the LVLm correctly describes the banana as black, it fails to do so when the image is distorted. As the level of visual uncertainty increases, the model increasingly relies on its language priors and favors more conventional banana colors such as “yellow” or “green”.

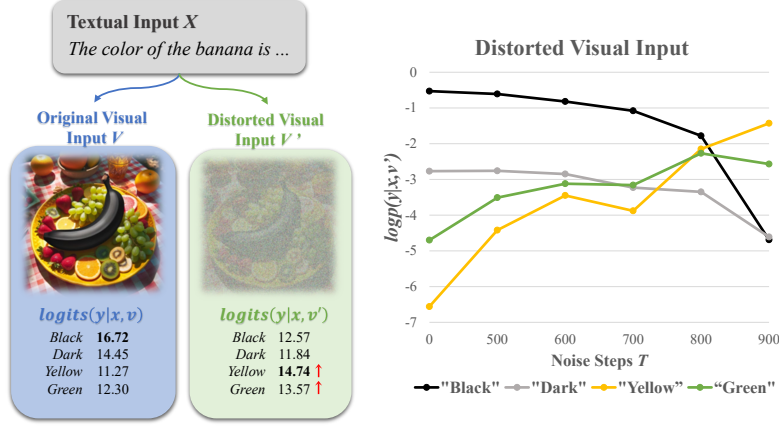


Figure 2.2: An illustration of visual uncertainty amplifying language priors [10].

A recent study performed a statistical analysis of key factors underlying object hallucinations: co-occurrence, uncertainty and text position. Uncertainty is the score used to measure a VLM’s confidence in generating the next token during decoding. Text position measures how early or late an object appears in generated text. As seen in Figure 2.3, object hallucinations tend to occur among objects that frequently co-occur, objects with a high uncertainty score during decoding and among objects that are generated in the later part of generated text [5]. Indeed, high co-occurrence and high uncertainty lead VLMs to hallucination due to over-reliance on statistical bias and language priors. Additionally, models might hallucinate more at the later parts of generated text due to the autoregressive text generation process. While the model starts off by relying on the semantic information present in the image, the information of past hallucinations and increasing uncertainty cause the model to “snowball”. Unlike initial hallucinations, “snowball hallucinations” occur when the model maintains consistency with past mistakes. [12].

2 Related Work

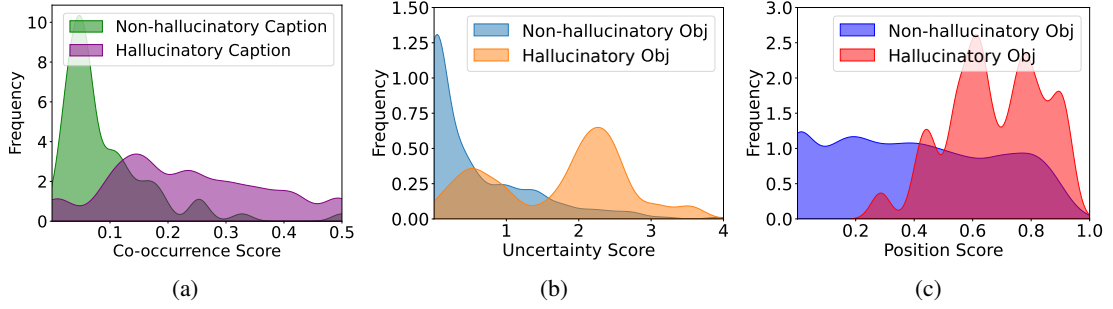


Figure 2.3: Comparison between hallucinatory and non-hallucinatory captions under different factors [8].

Hallucination Metrics

The standard way to measure the captioning performance of language models has been to measure the similarity of generated text to the corresponding ground truth caption. In practice, similarity based metrics are used to measure overall text quality, such as semantics and fluency. While these metrics do not directly probe hallucinations, an issue addressed by more recent hallucination-specific metrics, they still form the basis for evaluating text generation.

Stemming from the field of machine translation, BLEU [13] is an early attempt at automated text evaluation and still remains relevant today. BLEU is based on modified n-gram precision, which counts how many n-grams (phrases of 1 to 4 words) match a reference text. Matching single words measures the adequacy of generated text, while matching longer phrases measures fluency. BLEU combines these scores by taking the geometric mean of the modified n-gram precisions. As translations of the same text are expected to be similar in length, BLEU multiplies the n-gram precisions by a brevity penalty.

METEOR [14] was developed to address two issues in BLEU. First, BLEU’s n-gram matching is very rigid and only counts exact matches. In contrast, METEOR is more flexible and also matches synonyms and words sharing the same stem (e.g., “speaking” and “spoke”). Second, METEOR places more weight on unigram recall, the ratio of matched words to the total words in the reference text.

Adjacent to n-gram overlap metrics is ROUGE [15], a family of metrics focusing on different variations of n-gram recall. The recall based metric was developed for evaluating text summarization, consequentially it focuses on how much meaning is captured in comparison to reference summaries. ROUGE-N is the standard n-gram recall between a candidate text and a set of reference texts. Relying solely on n-grams can be rigid, one variant of ROUGE that addresses this issue is ROUGE-L, which measures the longest common subsequence between texts. For example the sequences “A B C D” and “A X B X C” do not share “A B C” as a strict 3-gram, but they do share “A B C” as their longest common subsequence. This has the advantage of capturing sentence structure and word order without needing a predefined n-gram length. ROUGE-L still has a blind spot, it weighs all subsequences equally, regardless of how spread out they are. For example, if the reference sequence is “A B C D E F”, ROUGE-L would give the

exact same score to the sequences “A B C X X X” and “A X X B X C”. ROUGE-W addresses this by weighing common subsequences by how spread out they are. ROUGE-S evaluates texts based on skip-bigrams, that is any word pairs in their sentence order allowing for arbitrary gaps. ROUGE-SU is a slightly less rigid version that also takes unigrams into account.

Generally, the goal in machine translation is accuracy and this is reflected in both BLEU and METEOR. Conversely, the main goal in image captioning is to generate human-like descriptions. BLEU has been shown to correlate weakly with human judgment, while METEOR showed a stronger correlation. CIDER [16] addresses this issue directly by evaluating generated text on a consensus based on multiple human written captions. Each n-gram is weighted according to the frequency that it appeared on different human written descriptions (Term Frequency), in other words, it places higher weight on words that multiple humans agreed were important enough to mention. Additionally, CIDER assumes that sequences that appear on every description in the dataset are less meaningful (e.g., “The image depicts”) and discounts these accordingly (Inverse Document Frequency). Based on these scores, CIDER finally compares the cosine similarity between the candidate description and the set of human reference sentences to calculate the final score. CIDER-D is a variation of CIDER designed to be more robust against the “gamification” of the metric. High CIDER scores can be achieved by repeating high confidence words over long sentences, CIDER-D remedies this by introducing a Gaussian penalty based on sentence length differences. Finally, CIDER-D also removes stemming to ensure the correct forms of words.

While the above metrics are based on n-gram overlap and recall, it’s also important to remember that n-gram overlap is neither necessary nor sufficient to compare the meaning of two texts. Two texts may use the same words but have different meanings or use completely different words and be semantically equivalent. SPICE [17] is a metric designed for image captioning that directly analyzes the semantic content of a given text. This is done by feeding the caption through a Probabilistic Context-Free Grammar (PCFG) dependency parser. The parser links words together based on their syntactic role in the sentence (e.g., Identifying “girl” as a central noun and drawing an adjectival modifier arrow from “young” to “girl”). This is used to build a semantic graph of the text. From this, tuples of word pairs (e.g., “(girl, young)”) are formed. The presence (or absence) of these tuples is compared to a reference text to calculate the final score. SPICE correlates better with human judgment than all the discussed n-gram overlap based metrics.

Modern image captioning models are able to write fluent and coherent image descriptions that might still contain objects that are not present in the scene. A generated caption that is rich and fluent but contains a singular hallucinated object might still score highly with text similarity based metrics, which is an obvious issue for sensitive applications. The CHAIR [18] metric tackles this issue directly by measuring the ratio of hallucinated to real objects. The two variants of CHAIR are $CHAIR_i$ which measures the ratio of hallucinated objects in a caption and $CHAIR_s$ which measures the ratio of sentences containing hallucinations. One advantage of CHAIR for image captioning is that it directly measures the rate of hallucinations within captions. On the flip-side, CHAIR inherits the sensitivity of language models to slight prompt variations. This means that 2 equivalent caption generation prompts might result in significantly different CHAIR scores (e.g., “Generate a short caption of the image.” vs. “Provide a brief description of the image.”).

2 Related Work

POPE [19] addresses this stability issue by reframing the evaluation of object hallucinations as a binary classification task. The POPE framework starts by generating 2 lists of objects, a list of ground-truth objects extracted from a given image and a list of objects not present in the image. Both lists are then used to generate binary probing questions (e.g., “Is there a bus in the image?”). One further advantage of POPE is the ability to create multiple benchmarks of varying difficulty by adjusting the list of nonexistent objects. For example, by selecting absent objects that often occur in the dataset or by selecting objects that co-occur often with the objects present in the image.

As the problem of hallucination evaluation is turned into a binary detection task, the standard metrics of signal detection theory become relevant. The main metrics used in binary hallucination evaluation are precision, recall, F-score and Yes Ratio. Assuming the case where the positive class is when an object is visible in the image, precision is the ratio of correctly predicted visible objects to the total predicted visible objects. This score is penalized when a model misses visible objects and incorrectly classifies them as not visible. Recall would then be the ratio of correctly predicted visible objects to the total actual visible objects. This score is penalized when a model is overly confident and classifies not visible objects as visible. The F_1 -score is the harmonic mean of precision and recall, this gives a good metric for model performance especially when the positive class is rare (if hallucinations are relatively rare, it might be sensible to use hallucination as the positive class). The F_β -score is the more general formulation allowing for different trade-offs between precision and recall (e.g., $\beta = 2$ gives more weight to recall over precision). One problem with the F-score is that it does not take into account the number of true negatives. Accuracy is the ratio of all correct predictions (both true positives and true negatives) to the total number of predictions, measuring the overall model performance. Finally, the Yes Ratio is the ratio of all positive predictions to the total number of predictions, measuring the model’s overall confidence and possible bias.

Hallucination Mitigation

Current research into hallucination mitigation spans across the entire lifecycle of VLMs. Modern approaches include explicit visual grounding and contextual enrichment of data, inference and decoding strategies that leverage the current understanding of hallucinations, and training models specifically for hallucination detection and revision.

One of the main sources of object hallucinations is over-reliance on learned language priors. For any domain, increasing the amount of data will not remove biases inherent in that domain. Thus, instead of increasing the amount of data, Biten et al. suggest augmenting textual training data with context irrelevant objects. For example, changing “a *bus* parked on a parking lot” into “a *banana* parked on a parking lot”. The introduced method consists broadly of three steps. The first step is a slight modification to the model architecture. Standard captioning models typically extract visual features from an image using a pretrained object detector. Specifically, an image I is processed by an object detector O to yield bounding box features V :

$$V = \{v_1, v_2, \dots, v_n\}$$

These features are then combined with an attention mechanism and fed into a language model in order to generate a sentence. To enable data augmentation, the authors concatenate embedded

object labels directly with the bounding box visual features. This extends the feature set from V to V' :

$$V' = \{v_1, v_2, \dots, v_n, l_1, l_2, \dots, l_n\}$$

The second step is to remove descriptive adjectives from sentences in order to prevent logical errors with inserted objects and to serve as a method of regularization. The final step is to replace the existing object labels with new labels. The authors introduce three sampling strategies. The simplest strategy is uniform sampling, where an object label is randomly selected from a uniform distribution. The second strategy is inverse multinomial sampling. Object labels are sampled from a distribution based on the co-occurrence matrix of all objects present in the dataset. Given an object o_i , an object o_j is more likely to be selected if their co-occurrence is low. Inverse multinomial sampling directly addresses bias issues regarding objects that frequently co-occur, it in turn creates a new bias for low co-occurrence objects. The third sampling strategy addresses this issue by continuously updating the co-occurrence matrix as each label is sampled. All strategies improve performance of state-of-the-art, with inverse multinomial sampling achieving the best performance for standard image captioning metrics while continuously updating the co-occurrence matrix achieving the best hallucination specific CHAIR score.

The method by Biten et al. utilizes data scrambling to prevent hallucinations that emerge due to over-reliance on language priors. A different factor that can result in hallucinations are datasets or data domains that are inherently more ambiguous. For example, the top-down perspective and large scale common in remote sensing satellite images can create difficult ambiguous scenarios for VLMs. For example there might not be enough visual information to correctly classify a beige rectangle in a city as a school, factory or clinic. In this case hallucinations might arise due to uncertainty caused by lack of information instead of stemming from data bias. The dataset fMoW-mm aims to improve remote sensing image captioning by enriching the input with map metadata, which might contain information not present in the raw satellite images. The authors of fMoW-mm achieve this by combining location, time and weather metadata of the fMoW-rgb dataset with the corresponding Open Street Map tiles to ultimately generate rich image captions using GPT-4o. This type of data augmentation is useful to augmenting the input and thus resulting in richer, more accurate image captions but also useful to train more robust models. The fMoW-mm dataset was used to train the CLIP model for object detection achieving better performance than the standard CLIP and even better performance than GeoRSCLIP and RemoteCLIP which were trained with significantly larger datasets.

While data centric hallucination mitigation methods are effective, they require data that might not be available or at least require model training which can be resource intensive. One alternative are methods that alter inference and token selection directly. One such method is FarSight, a decoding strategy developed to prevent two issues resulting from the attention matrix in transformer neural networks. The attention matrix is an $N \times N$ matrix where each cell (i, j) contains a weight representing how heavily token i is influenced by token j . In standard causal text generation, a model is strictly forbidden from looking at future tokens by masking the upper-triangular portion of the attention matrix (setting its values to negative infinity) before applying the softmax function, which turns those probabilities to 0. The first issue tackled by FarSight arises here. The use of the softmax function forces all (unmasked) attention scores to be non-zero,

2 Related Work

which causes the model to allocate disproportionately large attention to low-information tokens, such as visual backgrounds. The authors call this issue attention collapse. FarSight addresses this issue by exploiting the masked corner of the attention matrix and turning it into an attention register. By strategically placing values into this upper triangle before the softmax operation, FarSight gives the model a safe buffer to dump its excess, unneeded attention. This prevents the excess attention from collapsing onto irrelevant outlier tokens in the actual text. After the softmax operation, FarSight re-applies the causal mask in order to still prevent access to future tokens. The second issue addressed by FarSight is positional information decay, which is the progressive loss of attention to early tokens and therefore to the vision tokens. This happens because the rotational position encoding (RoPE) causes attention weight to decrease as the relative distance between tokens increases. The causal mask turned attention register is also used to solve this. Inside the attention register matrix P , FarSight applies a linear penalty based on a decay factor σ :

$$P_{i,j} = -(j - i) \cdot \sigma$$

Because the total attention across a row (valid tokens + register space) must equal exactly 1, manipulating the register directly changes the actual attention. Because the register’s absorption capacity decays for later rows (the progressively diminishing masking rate), the model is forced to allocate an increasingly larger share of its actual attention back to the valid tokens in the lower triangle. FarSight effectively decreases hallucinations produced by LLaVA-1.5, improving POPE and CHAIR scores.

One further method that targets inference directly is Visual Contrastive Decoding (VCD).

3 Proposed Method

This chapter describes the implementation of component X.

3.1 Basic Concepts

3.2 XXX

3.3 YYY

3.4 ZZZ

4 Data Set Description and Design of Experiments

4.1 Description of the Data Set

4.2 Design of Experiments

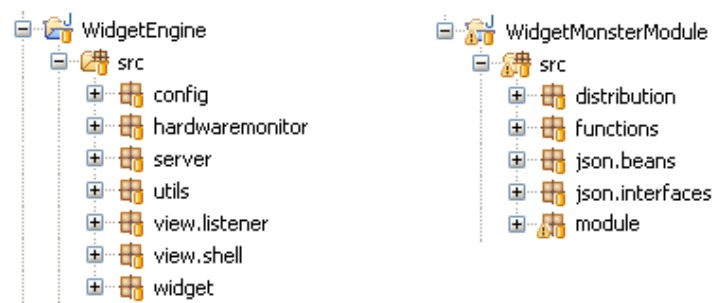


Figure 4.1: Project Structure Example

5 Experimental Results

This chapter describes the implementation of component X.

5.1 VLM hallucination detection performance

Are VLMs capable of detecting object hallucinations in Remote Sensing captions?

5.1.1 Binary questions for hallucination detection

Do binary questions on extracted objects successfully detect hallucinations?

Baseline performance

Table 5.1: Baseline performance of binary hallucination detection.

Dataset	Model	Model Size	Precision	Recall	F1-Score	Accuracy	Yes Ratio
LID	LLaVA 1.5	7B	0.5	0.5	0.5	0.5	0.5
	RS-LLaVA	7B	0.5	0.5	0.5	0.5	0.5
	Qwen3-VL	8B	0.5	0.5	0.5	0.5	0.5
LID	LLaVA 1.5	7B	0.5	0.5	0.5	0.5	0.5
	RS-LLaVA	7B	0.5	0.5	0.5	0.5	0.5
	Qwen3-VL	8B	0.5	0.5	0.5	0.5	0.5
LID	LLaVA 1.5	7B	0.5	0.5	0.5	0.5	0.5
	RS-LLaVA	7B	0.5	0.5	0.5	0.5	0.5
	Qwen3-VL	8B	0.5	0.5	0.5	0.5	0.5

Threshold optimization

Figure 5.1: 2x3 Violin plot and line plot for binary detection using LLaVA 1.5 for all 3 datasets.

5 Experimental Results

Figure 5.2: 2x3 Violin plot and line plot for binary detection using RS-LLaVA for all 3 datasets.

Figure 5.3: 2x3 Violin plot and line plot for binary detection using Qwen3-VL for all 3 datasets.

5.1.2 Binary questions vs Multiple Choice Questions

Are Multiple choice questions with multiple objects more suited for hallucination detection than single object binary questions?

5.1.3 Effect of VCD on hallucination detection using binary questions

Does VCD improve VLM hallucination detection?

Figure 5.4: Line plot for binary detection using VCD with the best performing model over α (alpha dictates the weight of VCD).

Effect of different distortion techniques on VCD performance

Are certain image distortions more suitable for performing VCD on Remote Sensing images?

Figure 5.5: Line plot of binary detection using VCD with varying levels of gaussian noise

Figure 5.6: Line plot of binary detection using VCD with varying levels of speckle

Figure 5.7: Line plot of binary detection using VCD with varying levels of masking

Figure 5.8: Line plot of binary detection using VCD with varying levels of masking

Table 5.2: Performance comparison of different masking strategies from ColorMAE

5.1.4 Semantic entropy performance

Is Semantic Entropy a viable strategy for rejecting hallucinations for Remote Sensing images?

Semantic entropy for VQA hallucination detection

Is Semantic Entropy a viable strategy for rejecting hallucinations for Remote Sensing VQA?

Table 5.3: Performance of semantic entropy for VQA questions of DDFAV

Semantic entropy for hallucination detection in captions

Is Semantic Entropy a viable strategy for rejecting hallucinations for Remote Sensing captions?

Table 5.4: Performance of semantic entropy for captions of dataset ???

5.2 New hallucination dataset

5.2.1 Detection performance difference between real and artificial hallucinations

Is Semantic Entropy a viable strategy for rejecting hallucinations for Remote Sensing captions?

5 *Experimental Results*

Table 5.5: Hallucination detection on LID (real tagged hallucinations) vs RSTeller (artificial hallucinations)

Table 5.6: Hallucination detection on LID (real tagged hallucinations) vs LID (artificial hallucinations)

Table 5.7: Hallucination detection on new dataset (real tagged hallucinations) vs new dataset (artificial hallucinations)

5.2.2 Corrected caption quality

Does a trained reviser create better quality corrected captions than zero-shot VLMs?

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Appendix